

QUANTUM MEASUREMENT

Projective measurements, observables, state update, generalized measurements, multi-qubit measurement, readout noise, and adaptive circuits

A detailed reference for quantum computing, quantum information, QKD, and experimental practice.

Topic 10 in the Quantum Mathematics Foundation series

How to use this volume

Quantum measurement is the bridge between a continuous complex state description and discrete classical data. This chapter separates three ideas that are often mixed together: the probability rule, the state-update rule, and the experimental sampling process. It develops projective measurement first, then extends the framework to mixed states, partial measurements, generalized measurements, noisy detectors, and adaptive circuits.

Learning objectives

- Calculate measurement probabilities from amplitudes, projectors, observables, and density matrices.
- Determine the conditional post-measurement state for projective and generalized measurements.
- Implement X-, Y-, and arbitrary-basis measurements using basis changes.
- Interpret expectation values and variances as statistics of repeated outcomes.
- Analyze partial measurements of entangled multi-qubit states.
- Distinguish projective measurements, measurement operators, and POVMs.
- Model finite-shot uncertainty and classical readout errors.
- Explain how mid-circuit measurement enables teleportation, error correction, and dynamic circuits.

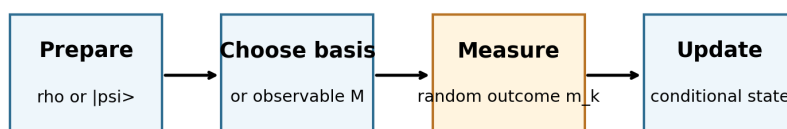
Contents

1	1. Why measurement is special
2	2. The measurement postulates
3	3. Computational-basis measurement
4	4. Basis measurements and basis changes
5	5. Projection operators
6	6. Observables and spectral decomposition
7	7. Expectation values and variance
8	8. Qubit measurement along an arbitrary axis
9	9. Bloch-sphere geometry of probability
10	10. Repeatability and state disturbance
11	11. Partial measurement of multi-qubit states
12	12. Entanglement and conditional states
13	13. Degenerate projective measurements
14	14. Density-matrix measurement rules
15	15. Measurement operators and Kraus form
16	16. POVMs and generalized measurements
17	17. Destructive and nondestructive detection
18	18. Incompatible observables and uncertainty
19	19. Information gain, disturbance, and no-cloning
20	20. Finite shots and statistical estimation
21	21. Readout errors and mitigation
22	22. Mid-circuit measurement, reset, and feed-forward
23	23. Bell-basis measurement
24	24. Measurement in QKD
25	25. Syndrome measurement in error correction
26	26. Measurement in Qiskit
27	27. Worked examples
28	28. Common mistakes
29	29. Exercises
30	30. Selected solutions
31	31. Compact reference sheet

1. Why measurement is special

Closed-system quantum evolution is linear and unitary. Measurement is different: it produces a classical outcome with a probability distribution and conditions the future state on that outcome. The state does not specify which individual result must occur; it specifies exact probabilities for all permitted outcomes.

Quantum measurement has probabilities and a conditional state change



The chosen measurement determines both the outcome labels and the state-update rule.

Measurement has three layers

Layer	Question	Mathematical object
Outcome structure	Which results are possible?	basis states, eigenspaces, or labels k
Probability rule	How likely is each result?	Born rule or $p(k) = \text{Tr}(\rho E_k)$
State update	What state remains after result k ?	projector P_k or measurement operator M_k

A fourth layer appears in experiments: a finite number of shots samples the theoretical distribution. Statistical fluctuations are not an additional quantum postulate; they are ordinary sampling noise after the Born probabilities have been fixed.

2. The measurement postulates

For an ideal projective measurement, let the mutually orthogonal projectors be $\{P_k\}$. They satisfy

$$P_k^\dagger = P_k, P_k P_j = \delta_{kj} P_k, \sum_k P_k = I.$$

If the pre-measurement pure state is $|\psi\rangle$, then the probability of result k is

$$p_k = \langle \psi | P_k | \psi \rangle = \|P_k |\psi\rangle\|^2.$$

If p_k is nonzero and result k occurs, the normalized conditional state is

$$|\psi_k\rangle = P_k |\psi\rangle / \sqrt{p_k}.$$

For a density matrix ρ , the corresponding rules are

$$p_k = \text{Tr}(P_k \rho), \rho_k = P_k \rho P_k / p_k.$$

These equations contain both the probability law and state update. The unconditional state after a measurement whose result is ignored is

$$\rho_{\text{after}} = \sum_k P_k \rho P_k.$$

Ignoring the result is different from conditioning on one result. The conditioned state can be pure even when the average post-measurement state is mixed.

3. Computational-basis measurement

For a qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, |\alpha|^2 + |\beta|^2 = 1.$$

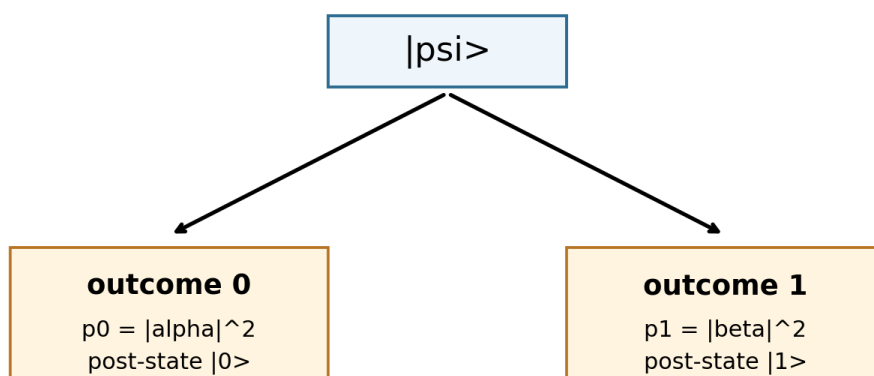
The computational-basis projectors are

$$P_0 = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, P_1 = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

The probabilities are

$$p(0) = \langle \psi | P_0 | \psi \rangle = |\alpha|^2, p(1) = |\beta|^2.$$

Projective measurement of $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$



The outcome is random, but the probability distribution is fixed by the state and measurement.

After outcome 0, the normalized state is $|0\rangle$; after outcome 1, it is $|1\rangle$. A computational-basis measurement therefore removes the relative phase and coherence between the two branches in the post-measurement ensemble.

Example

For $|\psi\rangle = (\sqrt{3}/2)|0\rangle + (i/2)|1\rangle$, the outcome probabilities are $3/4$ and $1/4$. The phase i affects interference in other bases but not the direct Z-basis probabilities.

4. Basis measurements and basis changes

A measurement basis is an orthonormal set $\{|b_k\rangle\}$. The probability of basis outcome k is

$$p_k = |\langle b_k | \psi \rangle|^2.$$

Hardware commonly measures in the computational basis. To measure in another basis, apply a unitary U that maps the desired basis vectors to computational basis vectors, then perform ordinary Z measurement.

A basis measurement can be implemented as a basis change plus Z measurement

Z-basis	X-basis	Y-basis
measure directly	apply H, then Z measure	apply $S^\dagger$ then H
$ 0\rangle, 1\rangle$	$ +\rangle, -\rangle$	$ +i\rangle, -i\rangle$

X basis

$$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}, \quad |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}.$$

Since $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$, applying H before Z measurement implements X measurement.

Y basis

$$|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}, \quad |-i\rangle = (|0\rangle - i|1\rangle)/\sqrt{2}.$$

Applying $S^\dagger$ and then H maps the Y eigenbasis to the computational basis. In circuit time order: first $S^\dagger$, then H , then measure Z .

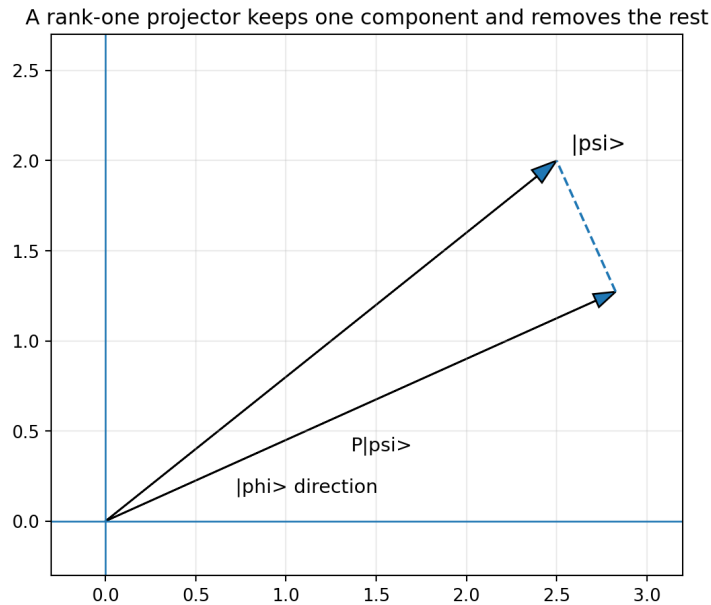
5. Projection operators

For a normalized vector $|\phi\rangle$, the rank-one projector onto its direction is

$$P_\phi = |\phi\rangle\langle\phi|.$$

It satisfies

$$P_\phi^\dagger = P_\phi, P_\phi^2 = P_\phi.$$



Acting on $|\psi\rangle$ gives

$$P_\phi|\psi\rangle = |\phi\rangle\langle\phi|\psi\rangle.$$

The scalar $\langle\phi|\psi\rangle$ is the overlap amplitude, and the squared norm of the projected component is the outcome probability.

Projector eigenvalues

Every projector has eigenvalues 0 and 1. Vectors inside the projected subspace are unchanged; orthogonal vectors are mapped to zero. This makes projectors natural yes/no observables.

Completeness

$$\sum_k |b_k\rangle\langle b_k| = I$$

for any complete orthonormal basis. Completeness guarantees that the measurement probabilities sum to one.

6. Observables and spectral decomposition

A standard quantum observable A is Hermitian. Its spectral decomposition is

$$A = \sum_k a_k P_k,$$

where a_k are real eigenvalues and P_k project onto the corresponding eigenspaces.

Measuring A means:

- 1 Obtain eigenvalue a_k with probability $p_k = \langle \psi | P_k | \psi \rangle$.
- 2 Update the state to $P_k | \psi \rangle / \sqrt{p_k}$ for a projective measurement.

For a nondegenerate observable, $P_k = |a_k\rangle\langle a_k|$. For a degenerate eigenvalue, P_k may have rank larger than one.

Example: Pauli Z

$$Z = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|.$$

A Z measurement returns mathematical values $+1$ or -1 . Software may store these outcomes as classical bits 0 or 1, but the observable values and bit labels should not be confused.

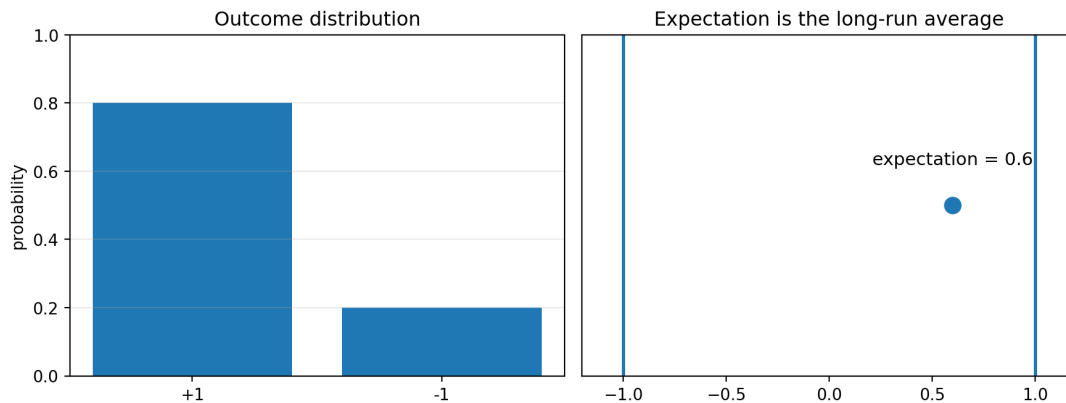
7. Expectation values and variance

The expectation value of observable A in state $|\psi\rangle$ is

$$\langle A \rangle = \langle \psi | A | \psi \rangle = \sum_k a_k p_k.$$

It is the average of many ideal measurement outcomes, not necessarily a possible result of one shot.

Individual outcomes can be ± 1 while the expectation lies between them



The variance is

$$\text{Var}(A) = \langle A^2 \rangle - \langle A \rangle^2.$$

For a Pauli observable, $A^2 = I$, so

$$\text{Var}(A) = 1 - \langle A \rangle^2.$$

Example

If a Z measurement has probabilities $p(+1)=0.8$ and $p(-1)=0.2$, then $\langle Z \rangle = 0.8 - 0.2 = 0.6$ and $\text{Var}(Z) = 1 - 0.36 = 0.64$.

Expectation values encode useful summaries, but a full probability distribution can contain information not visible in the mean alone.

8. Qubit measurement along an arbitrary axis

Any unit vector $n=(n_x, n_y, n_z)$ defines a Pauli observable

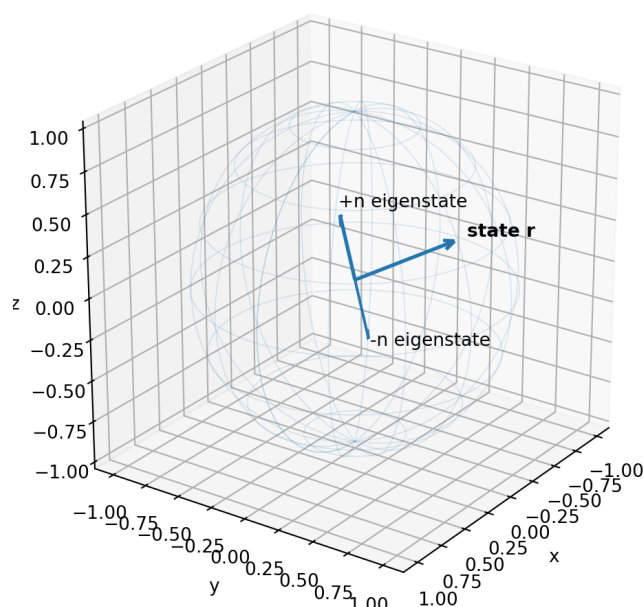
$$\sigma_n = n_x X + n_y Y + n_z Z.$$

Its eigenvalues are ± 1 , with eigenstates pointing along $\pm n$ on the Bloch sphere. If a qubit has Bloch vector r , then

$$p(+n) = (1 + r \cdot n)/2, \quad p(-n) = (1 - r \cdot n)/2,$$

$$\langle \sigma_n \rangle = r \cdot n.$$

A qubit measurement chooses an axis on the Bloch sphere



9. Bloch-sphere geometry of probability

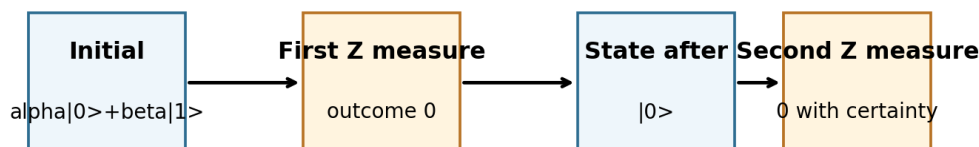
The dot product $r \cdot n$ measures alignment between the state and measurement axis. Parallel vectors give certainty for $+1$, antiparallel vectors give certainty for -1 , and perpendicular vectors give a 50/50 distribution.

Geometry	$r \cdot n$	$p(+n)$	$p(-n)$
parallel	1	1	0
perpendicular	0	1/2	1/2
antiparallel	-1	0	1

10. Repeatability and state disturbance

An ideal projective measurement is repeatable for the same observable: after obtaining eigenvalue a_k , the state lies in its eigenspace, so an immediate repetition returns the same value with certainty.

Repeatability of an ideal projective measurement



This repeatability is specific to ideal measurements of the same observable without intervening evolution.

However, an intervening unitary or an incompatible measurement can destroy that certainty. For example, measuring Z , then X , then Z generally makes the final Z outcome random.

Disturbance is basis dependent

Measuring a state already diagonal in the measurement basis may leave it unchanged. Measuring a coherent superposition in that basis destroys off-diagonal coherence in the unconditioned post-measurement density matrix.

$$\rho \rightarrow \sum_k P_k \rho P_k.$$

This map is often called dephasing in the measurement basis.

11. Partial measurement of multi-qubit states

To measure only subsystem A, use projectors P_k^A tensor I_B . For a joint state ρ_{AB} ,

$$p_k = \text{Tr}[(P_k^A \otimes I_B) \rho_{AB}].$$

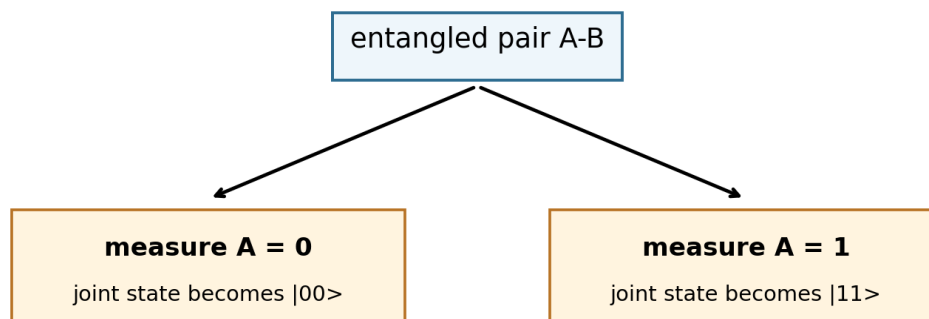
The conditional joint state is

$$\rho_{AB|k} = (P_k^A \otimes I) \rho_{AB} (P_k^A \otimes I) / p_k.$$

If subsystem B is kept, its conditional state is obtained with a partial trace over A.

12. Entanglement and conditional states

Partial measurement of the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$



A local outcome updates the conditional joint state and therefore predicts the remote qubit.

For the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, measuring qubit A in Z gives 0 or 1 with equal probability. Conditional on 0, the pair becomes $|00\rangle$; conditional on 1, it becomes $|11\rangle$. The reduced state of B before learning the result is maximally mixed:

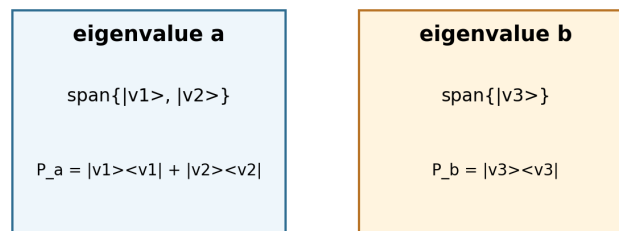
$$\rho_B = I/2.$$

No usable faster-than-light signal is created: without the classical outcome record, B remains described by the same reduced state.

13. Degenerate projective measurements

An observable is degenerate when several independent eigenvectors share one eigenvalue. The measurement outcome identifies the eigenspace but does not necessarily reveal which vector inside it was present.

Degenerate measurement: one outcome can correspond to a whole subspace



The measurement identifies the eigenspace, but it need not distinguish vectors inside that subspace.

For eigenvalue a with projector P_a , the probability and state update remain

$$p(a) = \langle \psi | P_a | \psi \rangle, \quad |\psi_a\rangle = P_a | \psi \rangle / \sqrt{p(a)}.$$

The projection preserves coherent superpositions inside the selected eigenspace. This distinction is important for multi-qubit observables such as parity. Measuring Z tensor Z distinguishes even parity from odd parity without determining the exact computational basis state within each parity sector.

Example: parity projectors

$$P_{\text{even}} = |00\rangle\langle 00| + |11\rangle\langle 11|, \quad P_{\text{odd}} = |01\rangle\langle 01| + |10\rangle\langle 10|.$$

14. Density-matrix measurement rules

Density matrices handle classical mixtures, subsystems of entangled states, and measurements whose outcome is ignored. For a projective measurement $\{P_k\}$:

$$p_k = \text{Tr}(P_k \rho).$$

$$\rho_k = P_k \rho P_k / p_k.$$

$$\rho_{\text{unconditioned}} = \sum_k P_k \rho P_k.$$

Example: dephasing a qubit

Let

$$\rho = \begin{bmatrix} |\alpha|^2 & \alpha \beta^* \\ \alpha^* \beta & |\beta|^2 \end{bmatrix}.$$

After a Z measurement whose result is discarded,

$$\rho_{\text{after}} = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}.$$

The diagonal probabilities remain, but the off-diagonal coherence disappears.

Selective versus nonselective measurement

Case	Outcome record	State description
Selective	condition on k	ρ_k
Nonselective	record ignored	$\sum_k p_k \rho_k$

15. Measurement operators and Kraus form

A generalized measurement can be specified by measurement operators $\{M_k\}$ satisfying

$$\sum_k M_k^\dagger M_k = I.$$

The probability of result k is

$$p_k = \langle \psi | M_k^\dagger M_k | \psi \rangle = \text{Tr}(M_k \rho M_k^\dagger).$$

The conditional state is

$$|\psi_k\rangle = M_k |\psi\rangle / \sqrt{p_k},$$

or for density matrices

$$\rho_k = M_k \rho M_k^\dagger / p_k.$$

The operators M_k describe both the statistics and the disturbance. Two different measurement implementations can have the same outcome probabilities but different post-measurement states.

Projective measurement is a special case

Set $M_k = P_k$. Then $M_k^\dagger M_k = P_k$ and the general formulas reduce to the projective rules.

16. POVMs and generalized measurements

The positive operator-valued measure (POVM) effects are

$$E_k = M_k^\dagger M_k.$$

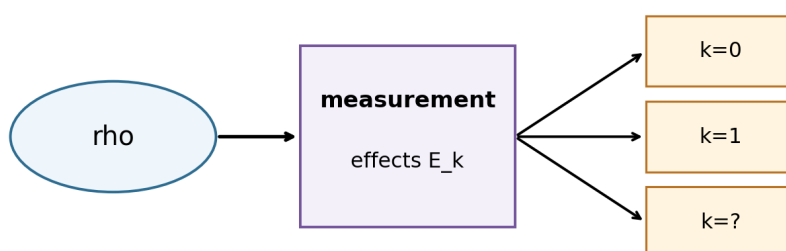
They satisfy

$$E_k \geq 0, \sum_k E_k = I,$$

and determine outcome probabilities through

$$p_k = \text{Tr}(\rho E_k).$$

POVMs describe generalized measurements and imperfect detectors



$$p(k) = \text{Tr}(\rho E_k), \quad E_k \geq 0, \quad \sum_k E_k = I$$

A POVM may have more outcomes than the Hilbert-space dimension, include an inconclusive outcome, or describe detector inefficiency. The effects E_k determine statistics but do not uniquely determine the state update; an instrument or a set of M_k is needed for that.

Example: unambiguous discrimination

Two nonorthogonal states cannot be perfectly distinguished every time. A POVM can instead return state 1, state 2, or an inconclusive result, with no erroneous conclusive identifications.

17. Destructive and nondestructive detection

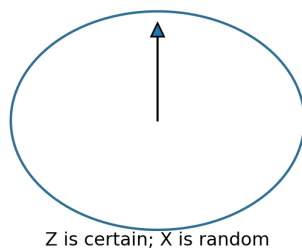
A destructive detector consumes or irreversibly changes the physical carrier, as in many photodetectors. A quantum nondemolition measurement aims to reveal an observable while preserving its eigenvalue for later checks. Nondemolition does not mean no disturbance at all; conjugate observables can still be disturbed.

18. Incompatible observables and uncertainty

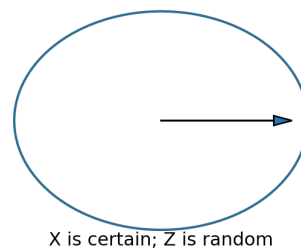
Observables A and B are incompatible when $[A,B]$ is nonzero. A state can be certain for one observable while necessarily uncertain for another.

Certainty in one incompatible basis can imply uncertainty in another

Z eigenstate $|0\rangle$



X eigenstate $|+\rangle$



The Robertson relation is

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A,B] \rangle|.$$

For Pauli operators, $[X,Z] = -2iY$ up to ordering convention, and the Bloch sphere makes the tradeoff geometric.

19. Information gain, disturbance, and no-cloning

If an unknown quantum state could be copied perfectly, one could measure different copies in incompatible bases and recover more information than quantum mechanics permits. The no-cloning theorem and measurement disturbance are therefore closely related limits on extracting information from unknown nonorthogonal states.

Orthogonal states can be perfectly distinguished and copied in a known basis; arbitrary nonorthogonal states cannot.

20. Finite shots and statistical estimation

A quantum circuit defines theoretical probabilities. Running it N times produces counts sampled from those probabilities. For a binary outcome with true probability p , the count K is binomial:

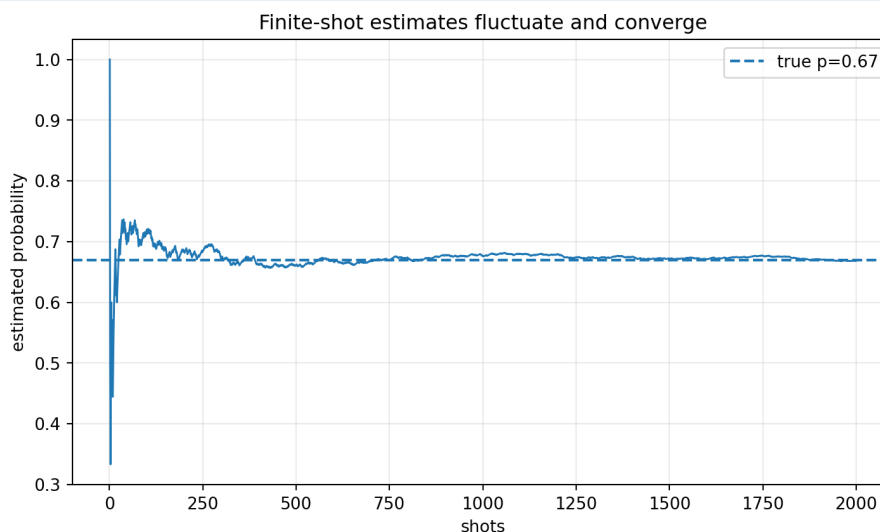
$$P(K=k)=C(N,k) p^k (1-p)^{(N-k)}.$$

The frequency estimator is

$$p_{\text{hat}}=K/N,$$

with standard error approximately

$$SE(p_{\text{hat}})=\sqrt{p(1-p)/N}.$$



To reduce standard error by a factor of 2, approximately 4 times as many shots are required. Finite shots do not reveal exact amplitudes; they estimate basis-dependent probabilities or expectation values.

Pauli expectation estimator

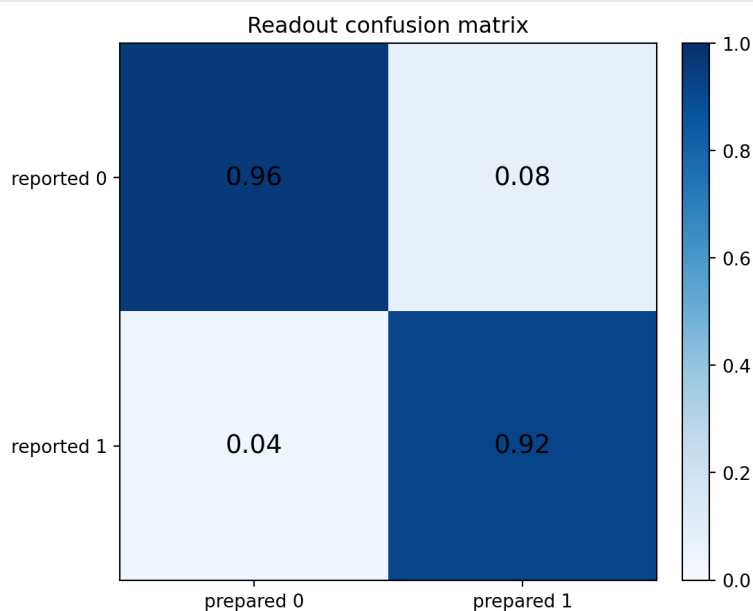
$$\langle P \rangle_{\text{hat}}=(n_{+} - n_{-})/N.$$

For independent shots, its variance is approximately $(1-\langle P \rangle^2)/N$.

21. Readout errors and mitigation

Real detectors can report the wrong classical bit. A simple classical model uses a confusion matrix M :

$$p_{\text{observed}} = M p_{\text{true}}.$$



The column j describes the distribution of reported outcomes when state j was prepared. If M is calibrated and well conditioned, one can estimate

$$p_{\text{true}} \text{ approximately } M^{-1} p_{\text{observed}}.$$

Cautions

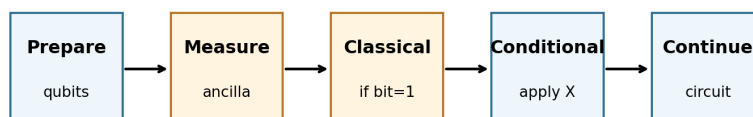
- Matrix inversion can amplify statistical noise.
- Readout errors may drift over time or depend on neighboring qubits.
- Mitigation reduces bias but does not restore the physical pre-measurement state.
- Negative corrected probabilities may appear and require constrained estimation.

Readout noise is classical after the quantum state reaches the detector, although the full hardware process can also include pre-measurement quantum errors.

22. Mid-circuit measurement, reset, and feed-forward

A measurement need not occur only at the end of a circuit. Mid-circuit measurement converts selected quantum information into a classical bit while other qubits remain active.

Mid-circuit measurement enables feed-forward and adaptive quantum protocols



The measurement outcome becomes classical data that selects a later operation.

Important operations

Operation	Purpose
Mid-circuit measurement	extract an intermediate classical condition
Reset	prepare a measured qubit again, often in $ 0\rangle$
Classical feed-forward	select a later gate based on a result
Postselection	retain only shots with chosen outcomes

Applications include teleportation corrections, iterative phase estimation, active reset, entanglement swapping, syndrome extraction, and adaptive error correction.

Timing matters: feedback latency must be short compared with the remaining qubits coherence time.

23. Bell-basis measurement

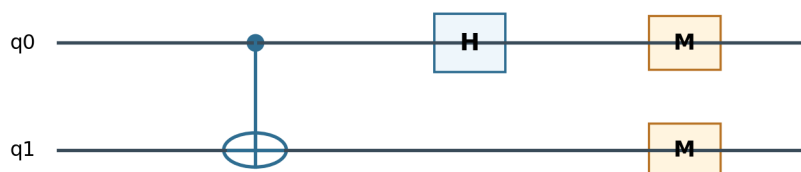
The Bell basis contains four orthonormal entangled states:

$$|\text{Phi}+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}, \quad |\text{Phi}-\rangle = (|00\rangle - |11\rangle)/\sqrt{2},$$

$$|\text{Psi}+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}, \quad |\text{Psi}-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}.$$

A CNOT followed by H on the first qubit maps these Bell states to distinct computational basis states.

Bell-basis measurement using CNOT, H, and computational measurement



The four Bell states are mapped to four distinct two-bit outcomes.

Bell input	after inverse preparation	measured bits
$ \text{Phi}+\rangle$	$ 00\rangle$	00
$ \text{Psi}+\rangle$	$ 01\rangle$	01
$ \text{Phi}-\rangle$	$ 10\rangle$	10
$ \text{Psi}-\rangle$	$ 11\rangle$	11

Bell measurements are central to teleportation, superdense coding, entanglement swapping, and some quantum-repeater protocols.

24. Measurement in quantum key distribution

In BB84, sender and receiver randomly choose between incompatible Z and X bases. A state prepared in one basis is deterministic when measured in that same basis and random when measured in the other.

BB84 uses two incompatible measurement bases

Basis	bit 0	bit 1	correct-basis result
Z	$ 0\rangle$	$ 1\rangle$	deterministic
X	$ +\rangle$	$ -\rangle$	deterministic
wrong basis	either state	either detector	random 50/50

An eavesdropper who guesses the wrong basis introduces detectable errors.

After publicly comparing basis choices, they keep only matched-basis positions. An eavesdropper who measures and resends without knowing the basis introduces errors because wrong-basis measurements disturb the state.

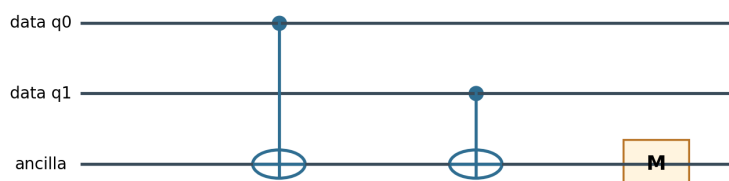
Measurement statistics used in QKD

- Quantum bit error rate (QBER) estimates disturbance and implementation noise.
- Different basis subsets can be used for key generation and parameter estimation.
- Finite-key security requires confidence bounds, not only observed frequencies.
- Detector mismatch and side channels require protocol-specific countermeasures beyond the ideal qubit model.

25. Syndrome measurement in quantum error correction

Quantum error correction measures stabilizer observables rather than the logical quantum information directly. Ancilla qubits interact with data qubits and are then measured.

Syndrome measurement extracts parity without directly reading the logical data



The ancilla outcome reveals even or odd parity and can guide error correction.

For a two-qubit Z parity check, the observable is Z tensor Z. Its eigenvalue is +1 on even-parity states $|00\rangle, |11\rangle$ and -1 on odd-parity states $|01\rangle, |10\rangle$.

The syndrome can reveal that an error occurred without revealing amplitudes inside the parity subspace. Repeated syndrome measurements over time help distinguish persistent data errors from faulty measurements.

This is a practical example of a degenerate, indirect, and often nondestructive measurement.

26. Measurement in Qiskit

A basic computational-basis measurement uses classical bits:

```
qc = QuantumCircuit(1, 1) qc.h(0) qc.measure(0, 0)
```

Running many shots returns counts such as {"0": 503, "1": 497}. The exact counts fluctuate, while the ideal probabilities are $1/2$ and $1/2$.

X-basis measurement

```
qc.h(0) # basis change qc.measure(0, 0)
```

Y-basis measurement

```
qc.sdg(0) qc.h(0) qc.measure(0, 0)
```

Measuring selected qubits

```
qc.measure([0, 2], [0, 1])
```

Practical cautions

- Qiskit bit-string ordering can differ from the visual qubit order; inspect classical-register mapping carefully.
- Measurement operations are not unitary and can restrict optimization across them.
- Statevector simulators usually require measurement removal or separate probability calculations for deterministic amplitudes.
- Dynamic-circuit support depends on backend capabilities.

27. Worked examples

Example 1: X measurement of a general qubit

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,

$$\langle +|\psi\rangle = (\alpha + \beta)/\sqrt{2}, \quad \langle -|\psi\rangle = (\alpha - \beta)/\sqrt{2}.$$

$$p(+)=|\alpha+\beta|^2/2, \quad p(-)=|\alpha-\beta|^2/2.$$

The relative phase between α and β changes these probabilities.

Example 2: arbitrary-axis qubit measurement

If $r=(0.6,0,0.8)$ and $n=(0,0,1)$, then $r \cdot n=0.8$, so

$$p(+z)=0.9, \quad p(-z)=0.1, \quad \langle Z \rangle = 0.8.$$

Example 3: unconditioned Z measurement

For $|+\rangle\langle+| = (1/2)[[1,1],[1,1]]$, ignoring the Z outcome gives

$$\rho_{\text{after}} = (1/2)[[1,0],[0,1]] = I/2.$$

The pure coherent state becomes a classical mixture in the Z basis.

Worked examples continued

Example 4: partial Bell measurement

For $|\Phi^+\rangle$, measuring the first qubit gives $p(0)=p(1)=1/2$. The conditional second-qubit state is $|0\rangle$ or $|1\rangle$, respectively. Averaging over unknown outcomes gives $I/2$.

Example 5: POVM probability

Let $E_0=(3/4)|0\rangle\langle 0|+(1/4)|1\rangle\langle 1|$ and $E_1=I-E_0$. For $\rho=|+\rangle\langle +|$,

$$p(0)=\text{Tr}(\rho E_0)=1/2, \quad p(1)=1/2.$$

These effects define statistics but not a unique state update.

Example 6: readout correction

Let $M=[[0.95,0.10],[0.05,0.90]]$ and $p_{\text{true}}=(0.8,0.2)^T$. Then

$$p_{\text{observed}}=M p_{\text{true}}=(0.78,0.22)^T.$$

A calibrated inverse can estimate the true distribution, but finite-shot noise may be amplified.

28. Common mistakes

Mistake	Correction
Treating measurement as merely reading a pre-existing classical value	Time is pre-empting classical value. Time jointly determine probabilities; incompatible measurements
Using amplitudes directly as probabilities	Squares squared magnitudes or use $\text{Tr}(\rho E_k)$.
Forgetting normalization after projection	Divide $P_k \psi\rangle$ by $\sqrt{p_k}$.
Assuming the expectation value is one's only output	One's only output is an eigenvalue; the expectation is an ensemble average.
Confusing a POVM effect with a state update operation	E_k determines probability; M_k determines both probability and disturbance.
Ignoring unrecorded outcomes	Discarding the record produces an averaged state, often with lost coherence.
Assuming a local Bell measurement sends information instantly	Conditional states on stage, but the remote reduced state without the record does not.
Ignoring classical-bit ordering in software	Trace qubit-to-classical-bit mapping explicitly.
Treating counts as exact probabilities	Counts have finite-shot uncertainty and hardware bias.

A useful habit

For every measurement problem, write four lines: measurement operators or projectors; possible labels or eigenvalues; outcome probabilities; conditional post-measurement states. This prevents most conceptual errors.

29. Exercises

Projective measurement

- 1 For $|\psi\rangle = (\sqrt{3}/2)|0\rangle + (i/2)|1\rangle$, calculate Z-basis probabilities and post-measurement states.
- 2 For $|\psi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$, derive X-basis probabilities.
- 3 Calculate the Y-basis probabilities of $|0\rangle$.
- 4 Show directly that $P_+ = |+\rangle\langle+|$ is Hermitian and idempotent.
- 5 For $A = 2|0\rangle\langle 0| - 3|1\rangle\langle 1|$ and $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, list possible outcomes and probabilities.
- 6 Find $\langle A \rangle$ and $\text{Var}(A)$ for the previous problem.

Multi-qubit measurement

- 1 Measure the first qubit of $(|00\rangle + |01\rangle + |10\rangle - |11\rangle)/2$ in the Z basis. Find probabilities and conditional states.
- 2 For $|\Phi^+\rangle$, find the probabilities of measuring the first qubit in the X basis and determine the conditional state of the second qubit.
- 3 Write the even- and odd-parity projectors for two qubits and verify that they sum to I .
- 4 Explain why parity measurement need not reveal which computational basis state is present.

Exercises continued

Generalized measurement and experiments

- 1 Given $M_0 = \text{diag}(1, \sqrt{1-\gamma})$ and $M_1 = \begin{bmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{bmatrix}$, verify completeness and find outcome probabilities for $\alpha|0\rangle + \beta|1\rangle$.
- 2 A POVM has $E_0 = (2/3)|0\rangle\langle 0| + (1/3)|1\rangle\langle 1|$ and $E_1 = I - E_0$. Find probabilities for $|+\rangle$.
- 3 A Z experiment gives counts $\{0: 720, 1: 280\}$. Estimate $p(0)$, $p(1)$, and $\langle Z \rangle$.
- 4 Estimate the standard error of $p(0)$ for $N=1000$ using the observed frequency.
- 5 A readout matrix is $\begin{bmatrix} 0.97 & 0.12 \\ 0.03 & 0.88 \end{bmatrix}$. Calculate the observed distribution for true probabilities $(0.6, 0.4)^T$.
- 6 Describe a basis-change circuit for measuring observable $(X+Z)/\sqrt{2}$ on one qubit.
- 7 Explain the roles of measurement, reset, and classical feed-forward in teleportation.
- 8 Why can a syndrome measurement reveal an error without measuring the encoded logical state directly?

Challenge problems

- 1 Derive $p(+n) = (1 + r \cdot n)/2$ using $\rho = (I + r \cdot \sigma)/2$ and $P_+ = (I + n \cdot \sigma)/2$.
- 2 Prove that an unconditioned projective measurement is trace preserving and completely positive.
- 3 Show that measuring one half of $|\Phi^+\rangle$ in any orthonormal qubit basis prepares the complex-conjugate basis state on the other half, up to the corresponding outcome label.

30. Selected solutions

Exercise 1

$$p(0)=3/4, p(1)=1/4.$$

The conditional states are $|0\rangle$ and $|1\rangle$.

Exercise 2

$$p(+)=|1+e^{i\phi}|^2/4=\cos^2(\phi/2),$$

$$p(-)=|1-e^{i\phi}|^2/4=\sin^2(\phi/2).$$

Exercise 3

Because $|\langle +|0\rangle|^2=|\langle -|0\rangle|^2=1/2$, the Y outcomes are equally likely.

Exercise 5

The eigenvalues are 2 and -3. Their probabilities are $|\alpha|^2$ and $|\beta|^2$.

Exercise 6

$$\langle A \rangle = 2|\alpha|^2 - 3|\beta|^2,$$

$$\langle A^2 \rangle = 4|\alpha|^2 + 9|\beta|^2, \text{ Var}(A) = \langle A^2 \rangle - \langle A \rangle^2.$$

Selected solutions continued

Exercise 7

Group by the first qubit:

$$|\psi\rangle = (1/2)|0\rangle(|0\rangle + |1\rangle) + (1/2)|1\rangle(|0\rangle - |1\rangle).$$

Each first-qubit outcome has probability $1/2$. The conditional second-qubit states are $|+\rangle$ and $|-\rangle$.

Exercise 8

Rewrite $|\Phi\rangle = (|++\rangle + |--\rangle)/\sqrt{2}$. The first-qubit X outcomes are equally likely and prepare $|+\rangle$ or $|-\rangle$ on the second qubit.

Exercise 11

The completeness relation is

$$M_0^\dagger M_0 + M_1^\dagger M_1 = I.$$

The probabilities are

$$p_0 = |\alpha|^2 + (1-\gamma)|\beta|^2, \quad p_1 = \gamma|\beta|^2.$$

Exercise 13

$$p_{\text{hat}}(0) = 0.72, \quad p_{\text{hat}}(1) = 0.28, \quad \langle Z \rangle_{\text{hat}} = 0.44.$$

Exercise 14

$$\text{SE} \approx \sqrt{0.72(0.28)/1000} \approx 0.0142.$$

Selected solutions continued

Exercise 15

$$p_{\text{observed}} = \begin{bmatrix} 0.97 & 0.12 \\ 0.03 & 0.88 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \end{bmatrix}^T = \begin{bmatrix} 0.63 & 0.37 \end{bmatrix}^T.$$

Challenge 1 sketch

Use

$$p(+n) = \text{Tr}(\rho P_+) = (1/4) \text{Tr}[(I + r \cdot \sigma)(I + n \cdot \sigma)].$$

The Pauli trace identities $\text{Tr}(\sigma_i) = 0$ and $\text{Tr}(\sigma_i \sigma_j) = 2 \delta_{ij}$ give

$$p(+n) = (1 + r \cdot n)/2.$$

Challenge 2 sketch

The channel $E(\rho) = \sum_k P_k \rho P_k$ has a Kraus representation with Kraus operators P_k , so it is completely positive. Trace preservation follows from

$$\text{Tr}[E(\rho)] = \text{Tr}[\rho \sum_k P_k^\dagger P_k] = \text{Tr}(\rho).$$

31. Compact reference sheet

Concept	Formula
Projective completeness	$\sum_k P_k = I, P_k P_j = \delta_{kj} P_k$
Pure-state probability	$p_k = \langle \psi P_k \psi \rangle$
Pure conditional state	$P_k \psi \rangle / \sqrt{p_k}$
Density-matrix probability	$p_k = \text{Tr}(P_k \rho)$
Density conditional state	$P_k \rho P_k / p_k$
Unconditioned projective update	$\sum_k P_k \rho P_k$
Basis probability	$ \langle b_k \psi \rangle ^2$
Observable decomposition	$A = \sum_k a_k P_k$
Expectation	$\langle A \rangle = \text{Tr}(\rho A)$
Variance	$\text{Tr}(\rho A^2) - \langle A \rangle^2$
Arbitrary-axis qubit probability	$p(+n) = (1 + \langle \mathbf{r} \cdot \mathbf{n} \rangle) / 2$
General measurement completeness	$\sum_k M_k^\dagger M_k = I$
General probability	$p_k = \text{Tr}(M_k \rho M_k^\dagger)$
General conditional state	$M_k \rho M_k^\dagger / p_k$
POVM effect	$E_k = M_k^\dagger M_k$
POVM probability	$p_k = \text{Tr}(\rho E_k)$
Binary frequency estimator	$\hat{p} = K/N$
Frequency standard error	$\sqrt{p(1-p)/N}$
Pauli expectation estimate	$(n_+ - n_-) / N$
Readout model	$p_{\text{observed}} = M p_{\text{true}}$

Final perspective

Quantum measurement is not a single formula. It is a structured transformation from quantum states to classical outcomes, with a probability law, conditional state update, and an experimental sampling process. Projective measurements provide the clean foundation; density matrices, Kraus operators, and POVMs extend it to realistic detectors, noisy processes, partial information, and adaptive protocols.

The next topic in the roadmap is single-qubit gates, followed by Hadamard interference and the Bloch sphere in greater operational detail.